

Load data

```
In [4]: import pandas as pd

df = pd.read_csv('../data/cleaned_hr_data.csv')
```

EDA

```
In [5]: df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15002 entries, 0 to 15001
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   satisfaction_level     15002 non-null  float64
 1   last_evaluation        15002 non-null  float64
 2   number_project         15002 non-null  int64  
 3   average_monthly_hours  15002 non-null  int64  
 4   time_spend_company     15002 non-null  int64  
 5   Work_accident          15002 non-null  int64  
 6   employee_left          15002 non-null  int64  
 7   promotion_last_5years  15002 non-null  int64  
 8   department             15002 non-null  str     
 9   salary                 15002 non-null  str     
dtypes: float64(2), int64(6), str(2)
memory usage: 1.3 MB
```

```
Out[5]:
```

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spend_company
--	--------------------	-----------------	----------------	-----------------------	--------------------

count	15002.000000	15002.000000	15002.000000	15002.000000	15002.000000
mean	0.612839	0.716068	3.802693	201.041728	2.806358
std	0.248607	0.171179	1.232733	49.941815	1.453991
min	0.090000	0.360000	2.000000	96.000000	1.000000
25%	0.440000	0.560000	3.000000	156.000000	1.000000
50%	0.640000	0.720000	4.000000	200.000000	2.000000
75%	0.820000	0.870000	5.000000	245.000000	3.000000
max	1.000000	1.000000	7.000000	310.000000	4.000000



```
In [6]: df['employee_left'].value_counts()
```

```
Out[6]: employee_left
0      11428
1       3574
Name: count, dtype: int64
```

Heatmap

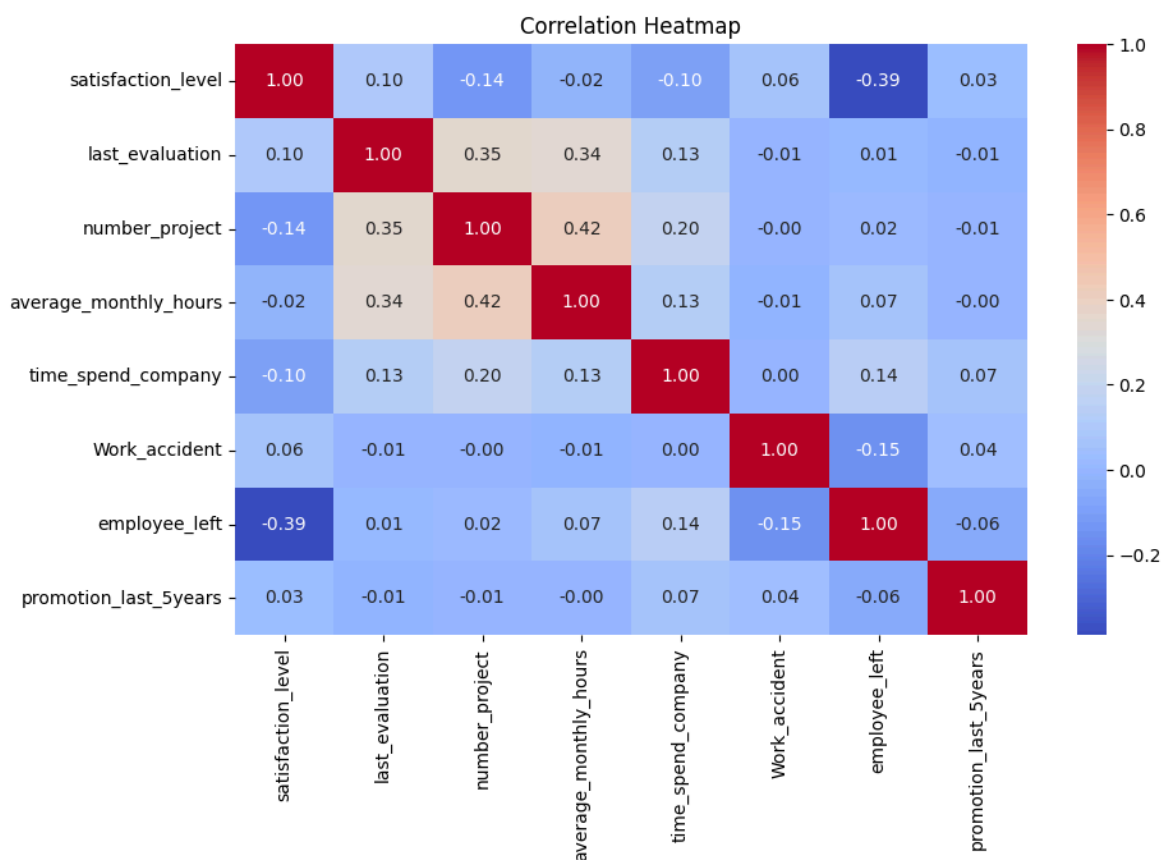
```
In [7]: import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))

corr = df.select_dtypes(include=['int64', 'float64']).corr()

sns.heatmap(
    corr,
    annot=True,
    fmt='.2f',
    cmap='coolwarm'
)

plt.title("Correlation Heatmap")
plt.show()
```



Correlation Heatmap Insights

Satisfaction Level shows the strongest negative correlation with attrition (-0.39), indicating that employees with lower satisfaction are significantly more likely to leave.

Time Spent at Company has a slight positive relationship with attrition (0.14), suggesting attrition increases with tenure up to a certain point.

Work Accidents are negatively correlated with attrition (-0.15), meaning employees involved in workplace incidents were less likely to leave.

Number of Projects and Monthly Hours are moderately correlated (0.42), indicating employees with more projects tend to work longer hours.

```
In [8]: # Satisfaction Level vs Attrition

plt.figure(figsize=(6, 6))

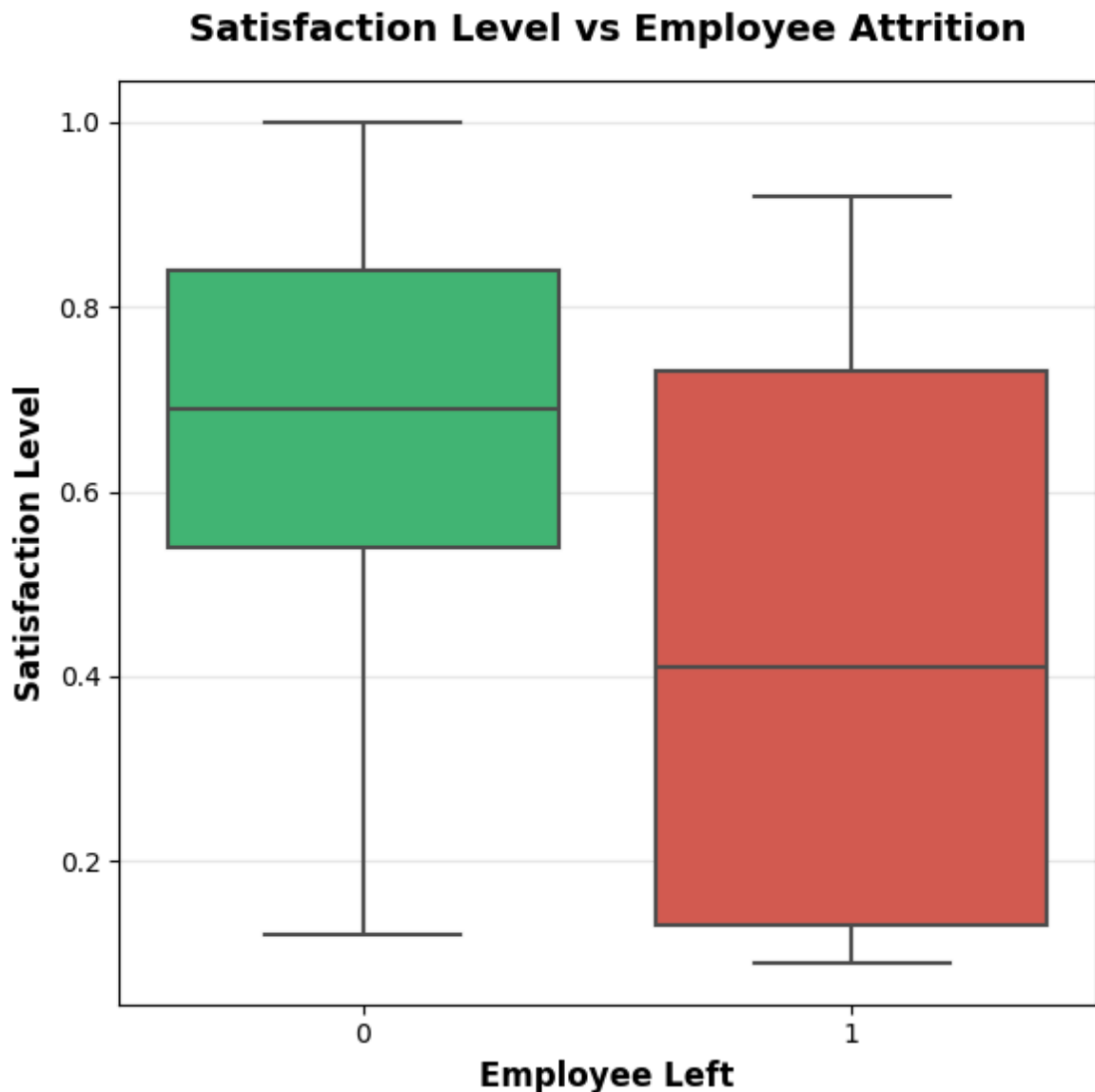
ax = sns.boxplot(
    data=df,
    x='employee_left',
    y='satisfaction_level',
    palette=['#2ecc71', '#e74c3c'],
    linewidth=1.5,
    fliersize=4,
    flierprops=dict(marker='o', markerfacecolor='gray', markersize=4, alpha=0.5)
)

# Customize title and labels
plt.title("Satisfaction Level vs Employee Attrition", fontsize=14, fontweight='bold')
plt.xlabel("Employee Left", fontsize=12, fontweight='semibold')
plt.ylabel("Satisfaction Level", fontsize=12, fontweight='semibold')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
```

C:\Users\wardo\AppData\Local\Temp\ipykernel_18396\3884130296.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v 0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
ax = sns.boxplot(
```



Satisfaction Level vs Employee Attrition Insights

Employees who left the company had significantly lower satisfaction levels compared to those who stayed.

The median satisfaction score of employees who left is substantially lower than that of retained employees.

Employee satisfaction appears to be one of the most important drivers of attrition.

In [9]: *# Department vs Attrition Rate*

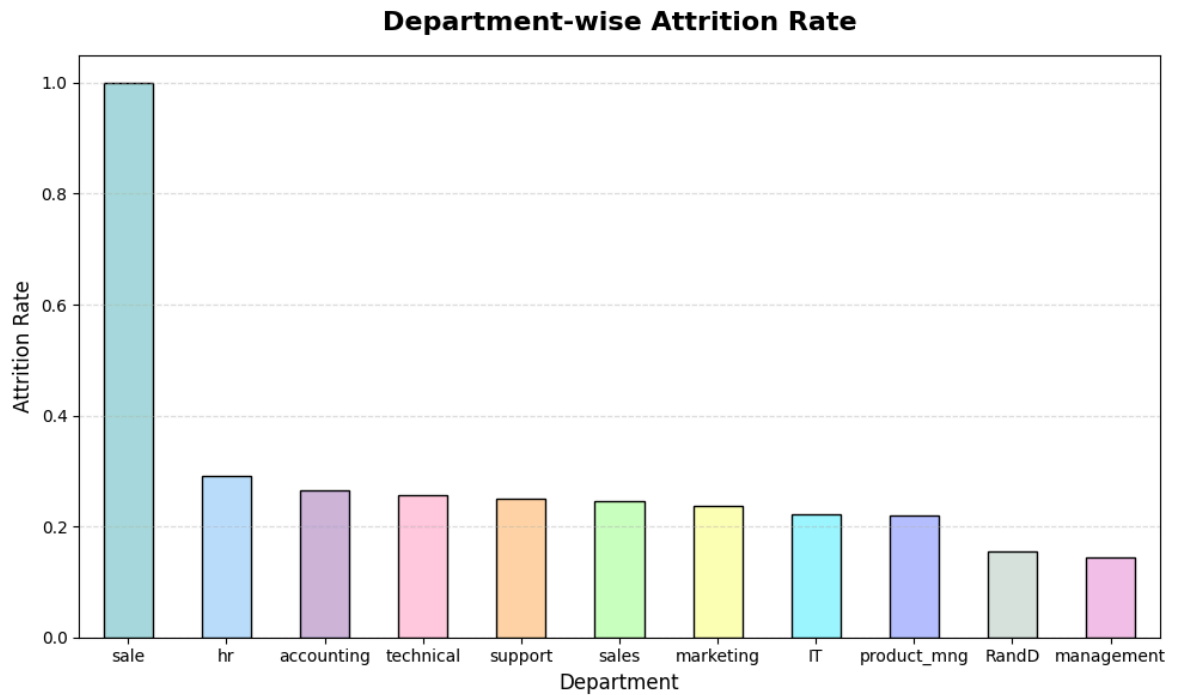
```
dept_attrition = df.groupby('department')['employee_left'].mean().sort_values(a
colors = ['#A8DADC', '#BDE0FE', '#CDB4DB', '#FFC8DD', '#FFD6A5', '#CAFFBF', '#FD
plt.figure(figsize=(10,6))

dept_attrition.plot(
    kind='bar',
    color=colors,
    edgecolor='black'
```

```

)
plt.title(
    "Department-wise Attrition Rate",
    fontsize=16,
    fontweight='bold',
    pad=15
)
plt.xlabel("Department", fontsize=12)
plt.ylabel("Attrition Rate", fontsize=12)
plt.xticks(rotation=0, fontsize=10)
plt.yticks(fontsize=10)
plt.grid(axis='y', linestyle='--', alpha=0.4)
plt.tight_layout()

```



Department-wise Attrition Rate Insights

HR recorded the highest attrition rate among major departments (~29%).

Accounting and Technical departments also experienced relatively high turnover.

Management and R&D showed the lowest attrition rates, indicating stronger employee retention.

The Sales department had a moderate attrition rate despite having the largest workforce.

```

In [10]: # Salary vs Attrition Rate

salary_attrition = df.groupby('salary')['employee_left'].mean()

colors = ['#A8DADC', '#FEBDF4', '#BDB4DB']

plt.figure(figsize=(8,6))
salary_attrition.plot(

```

```
kind='bar',  
color=colors,  
edgecolor='black',  
width=0.7  
)  
plt.title(  
    'Salary-wise Attrition Rate (%)',  
    fontsize=16,  
    fontweight='bold',  
    pad=15  
)  
plt.xlabel('Salary Level', fontsize=12)  
plt.ylabel('Attrition Rate (%)', fontsize=12)  
plt.xticks(rotation=0, fontsize=11)  
plt.yticks(fontsize=11)  
plt.grid(axis='y', linestyle='--', alpha=0.3)  
plt.tight_layout()
```



Salary-wise Attrition Rate Insights

- Employees in the low-salary category experienced the highest attrition rate (~30%).
- Attrition decreased significantly as salary levels increased.
- High-salary employees had the lowest turnover rate (~7%).
- Compensation appears to be a major factor influencing employee retention.

```
In [11]: # Number of Projects vs Attrition Rate

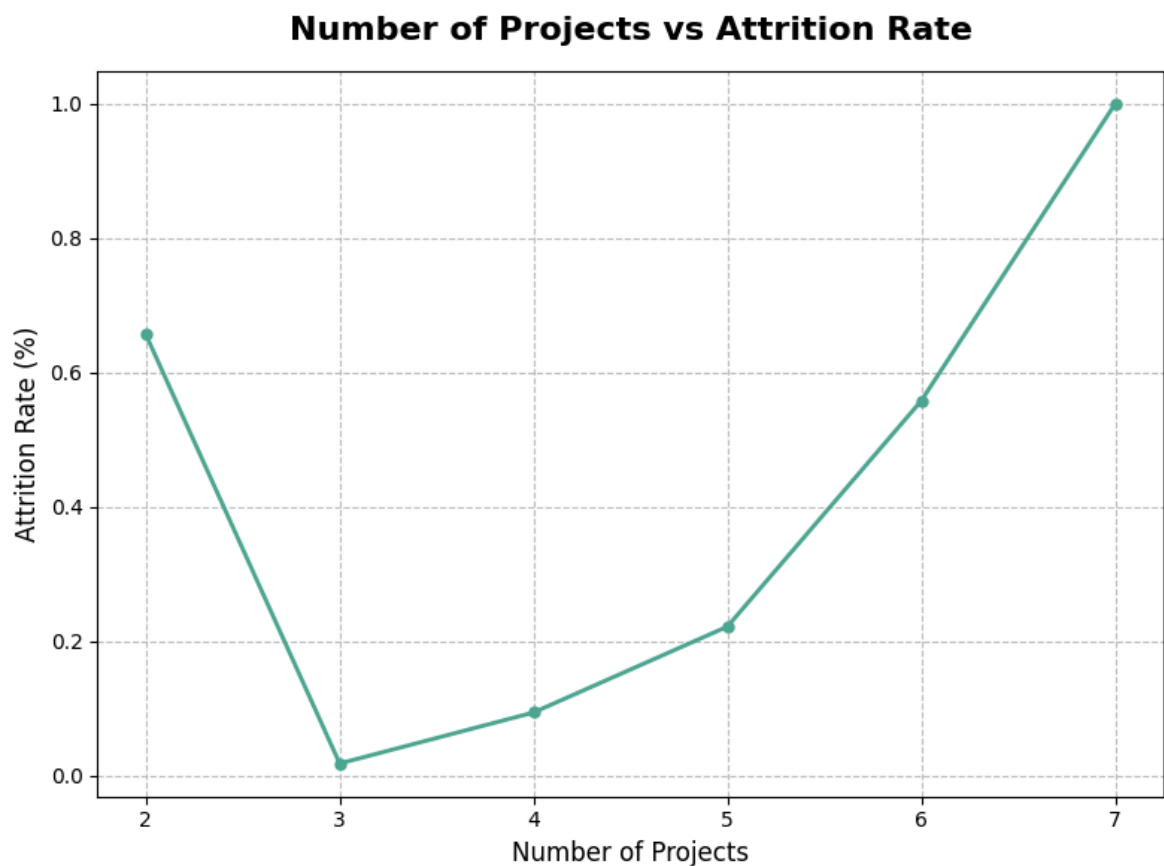
project_attrition = df.groupby('number_project')['employee_left'].mean()

plt.figure(figsize=(8,6))

project_attrition.plot(
    kind='line',
    marker='o',
    linewidth=2,
    markersize=5,
    color="#4EA792"
)

plt.title(
    "Number of Projects vs Attrition Rate",
    fontsize=16,
    fontweight='bold',
    pad=15
)

plt.xlabel("Number of Projects", fontsize=12)
plt.ylabel("Attrition Rate (%)", fontsize=12)
plt.xticks(fontsize=10)
plt.yticks(fontsize=10)
plt.grid(True, linestyle='--', alpha=0.8)
plt.tight_layout()
```



Number of Projects vs Attrition Rate Insights

Employees handling only 2 projects exhibited extremely high attrition rates.

Attrition remained relatively low for employees managing 3–5 projects.

Attrition increased sharply for employees assigned 6–7 projects.

The results suggest both underutilization and excessive workload contribute to employee turnover.

```
In [12]: # Average Monthly Hours vs Attrition Rate

hours_attrition = df.groupby('average_monthly_hours')['employee_left'].mean()

plt.figure(figsize=(13,6))

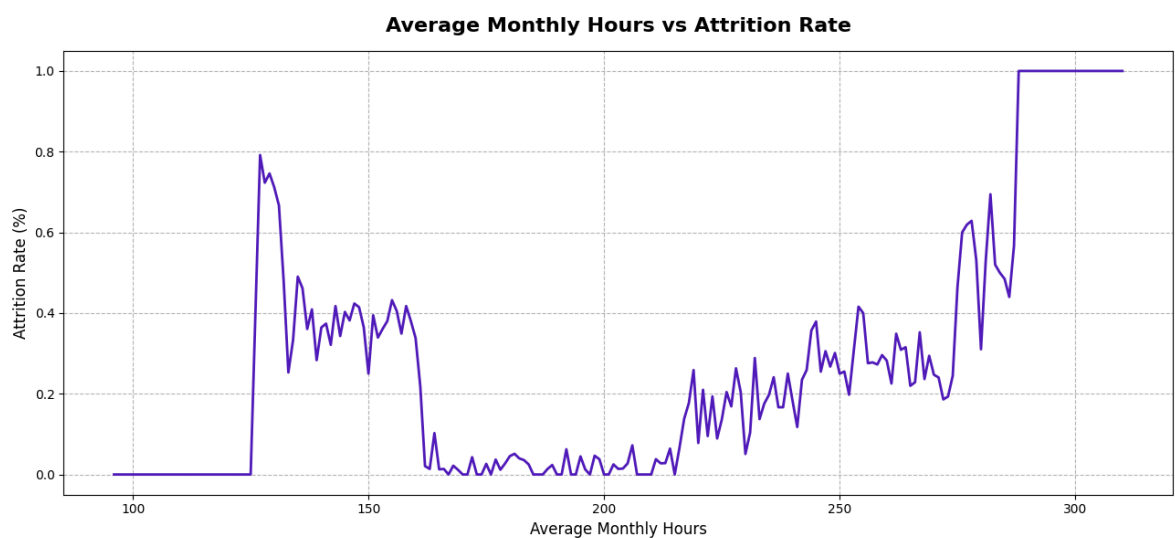
hours_attrition.plot(
    kind='line',
    color="#501BBA",
    linewidth=2
)

plt.title(
    'Average Monthly Hours vs Attrition Rate',
    fontsize=16,
    fontweight='bold',
    pad=15
)

plt.xlabel('Average Monthly Hours', fontsize=12)
plt.ylabel('Attrition Rate (%)', fontsize=12)

plt.grid(
    True,
    linestyle='--',
    alpha=0.9
)

plt.xticks(fontsize=10)
plt.yticks(fontsize=10)
plt.tight_layout()
```



Average Monthly Hours vs Attrition Rate Insights

Employees working very high monthly hours experienced significantly higher attrition rates.

Attrition remained relatively low for employees with moderate workloads.

The trend suggests employee burnout may be a major contributor to turnover.

Excessive working hours are associated with increased employee departure risk.

```
In [13]: # Time Spent in Company vs Attrition Rate

tenure_attrition = (df.groupby('time_spend_company')['employee_left'].mean())

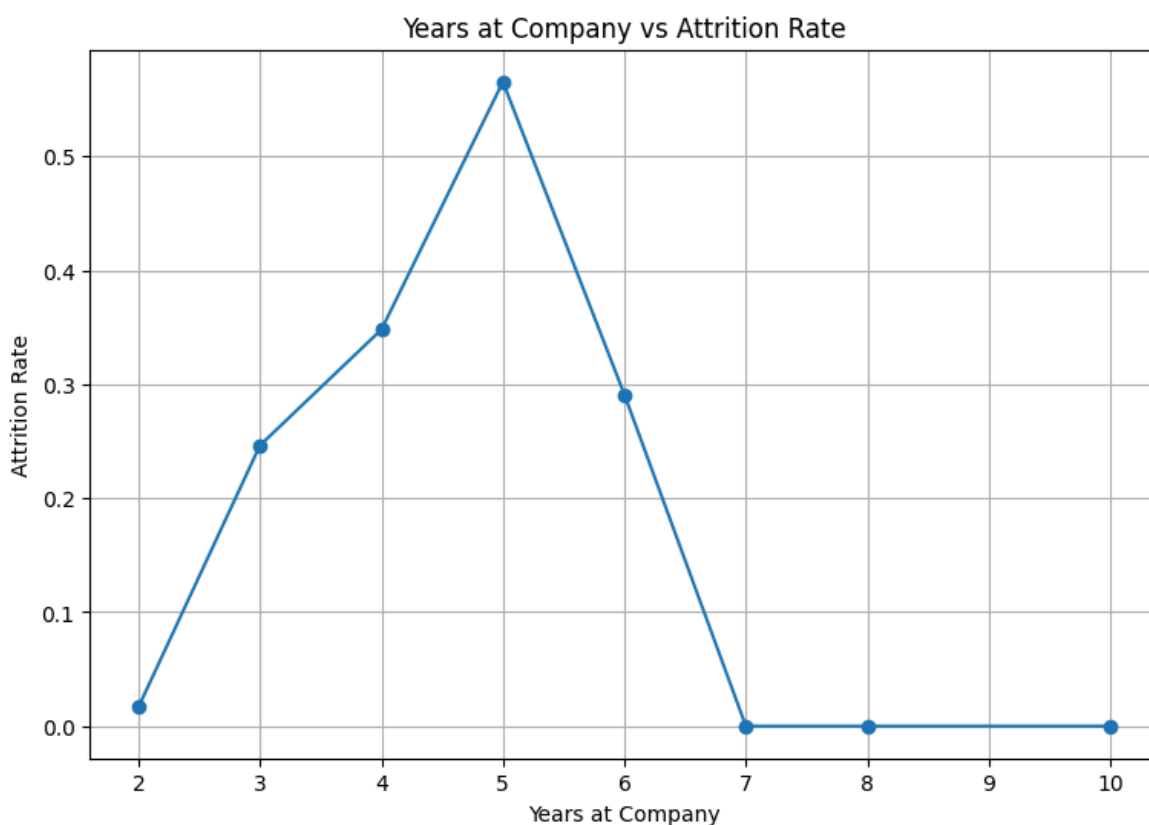
plt.figure(figsize=(9,6))

tenure_attrition.plot(
    kind='line',
    marker='o'
)

plt.title("Years at Company vs Attrition Rate")
plt.xlabel("Years at Company")
plt.ylabel("Attrition Rate")

plt.grid(True)

plt.show()
```



Years at Company vs Attrition Rate Insights

- Attrition increased steadily from 2 to 5 years of tenure.
- Employees with approximately 5 years at the company showed the highest attrition rate.
- After 6 years, attrition declined considerably.
- Mid-tenure employees appear to be the most vulnerable group for employee turnover.

Overall Business Insights

- Employee satisfaction is the strongest predictor of attrition.
- Low salary levels are strongly associated with higher employee turnover.
- Both underworked and overworked employees demonstrate elevated attrition rates.
- Employees with heavy workloads and long working hours are at greater risk of leaving.
- HR, Accounting, and Technical departments require targeted retention strategies due to higher turnover rates.
- Employees reaching the 4–5 year tenure range represent a critical retention segment.
- Promotion opportunities appear limited and may contribute indirectly to attrition.
- Maintaining competitive compensation, balanced workloads, and employee engagement programs can significantly improve retention outcomes.
- Workforce retention efforts should primarily focus on low-satisfaction, low-salary, and high-workload employees, as these groups exhibit the highest attrition risk.

```
In [14]: df.head()
```

Out[14]:

	satisfaction_level	last_evaluation	number_project	average_monthly_hours	time_spen
0	0.38	0.53	2	157	
1	0.80	0.86	5	262	
2	0.11	0.88	7	272	
3	0.72	0.87	5	223	
4	0.37	0.52	2	159	

